Measurement-Device-Independent Quantum Key Distribution

A Technical Review

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Background and Motivation

Why Quantum Key Distribution?

- Measuring a quantum system **disturbs** the state.
- Eavesdroppers introduce detectable **anomalies**.
- Exploit quantum physics to detect any eavesdropping.

$$\text{Alice} \xrightarrow{\psi} \text{Eve} \xrightarrow{\psi} \text{Bob}$$

Classical:
No disturbance

$$\text{Alice} \xrightarrow{\psi} \text{Eve} \xrightarrow{\psi'} \text{Bob}$$

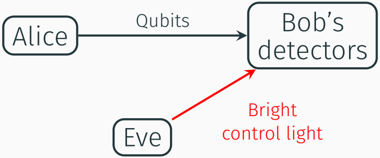
Quantum:
Disturbed state

Early QKD: BB84

- Alice encodes bits in random bases.
- Bob measures in random bases.
- Basis sifting and error tests reveal eavesdropping.

Shot	1	2	3	4	5	6	7	8	9	10
Alice bit	0	1	1	0	1	1	1	0	0	1
Alice basis	+	×	+	+	×	×	+	×	+	+
Bob basis	×	×	+	×	+	×	×	×	+	+
Bob outcome	1	1	1	0	0	1	1	1	0	1
Error test?	-	Y	N	-	-	N	-	Y	N	N
Keep bit?	N	Y	Y	N	N	Y	N	N	Y	Y
Shared secret key	-	1	1	-	-	1	-	-	0	1

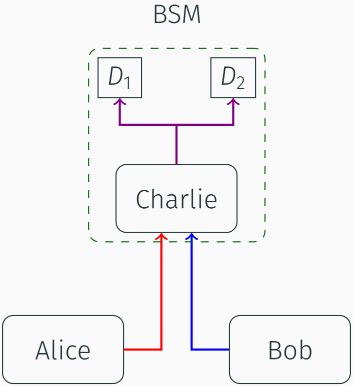
- BB84 security **assumes** ideal detectors.
- Real detectors have imperfections and control signals.
- Eve can exploit to **fake** valid detections.



Measurement-Device-Independence

The Innovation: Trustless Detectors

- Both parties send prepared states.
- Bit selection via **entanglement** measurements at a central relay.
- Security not dependent on detector behaviour.



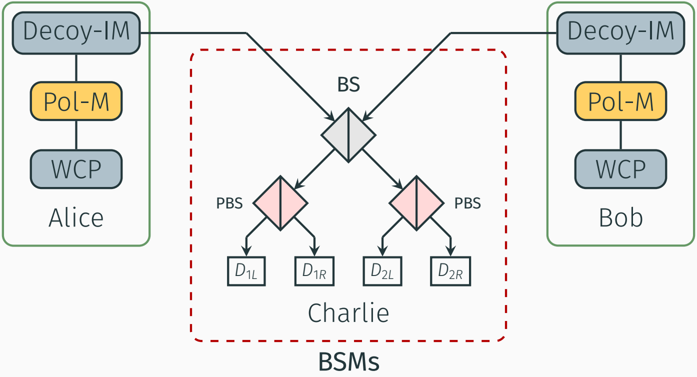
Why MDI?

- Detector vulnerabilities exposed **security limits** of BB84.
- MDI closes this gap: detectors become untrusted.
- Ongoing work: efficient sources, long-distance links, integration with networks.

Component	BB84	MDI
Sources	×	×
Fibres	✓	✓
Detectors	×	✓

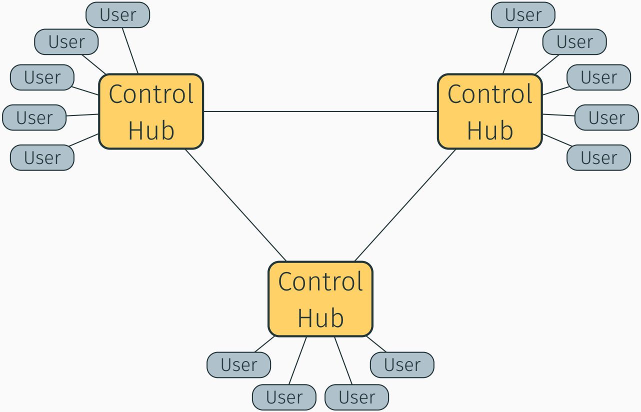
Standard Setup

- Time-synchronised weak coherent pulses (**WCPs**) from sources.
- Phase and intensity **modulators** encode bits and bases.
- Interference at relay allows Bell state measurements (**BSMs**).



Experimental Modifications

- Quantum-dot **single-photon sources** (Zhou et al. 2018: 100× key-rate improvement via blockades).
- High user-count star/ring-star topologies used.



Conclusion

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- Detector imperfections are a fundamental **vulnerability**.
- MDI **shifts trust**: sources trusted, detectors untrusted.
- Practical quantum networks now feasible **without trust** assumptions.